NARMADHA PALANISAMY

Senior Test Automation Engineer / SDET | API & UI Framework Development | Karate • REST Assured • Playwright

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PROFESSIONAL SUMMARY

Senior Test Automation Engineer with 15+ years in software and quality engineering who designs and builds automation frameworks from the ground up. Builds API automation with Karate, REST Assured and Playwright, and UI automation with Playwright (TypeScript) and Selenium (Java), with reusable components, parallel execution and Jenkins CI/CD quality gates that cut manual testing by 70%. Led the modernization of a legacy test suite into Playwright and validates data with SQL across relational databases. Hands-on with AWS (Lambda, S3, CloudWatch, Secrets Manager) and Docker-based test execution, and uses AI dev-assist tools (Claude, GitHub Copilot) daily for code generation, debugging and documentation. Takes full ownership from framework design through execution while partnering closely with developers.

TECHNICAL SKILLS

API Automation: Karate, REST Assured (Java), Playwright API testing, Postman/Newman; REST APIs, JSON/XML, JSON schema validation of service contracts, OAuth 2.0, JWT

UI Automation: Playwright (TypeScript/JavaScript), Selenium WebDriver (Java), Cucumber/BDD, Page Object Model, parallel execution

Frameworks & Design: Framework architecture, reusable test components, utilities and libraries, test data builders (Faker), fixtures, TestNG, JUnit, Pytest

Languages: Java (Oracle Certified Java Professional), TypeScript, JavaScript, Python, SQL, Bash

CI/CD & Containers: Jenkins (multibranch pipelines), GitHub Actions, GitLab, Git, Maven, Docker (containerized test execution)

Databases: SQL for validation and test setup: Oracle, MySQL, PostgreSQL, Snowflake; MongoDB; SOQL

Cloud: AWS Lambda, S3, CloudWatch, Secrets Manager, IAM; AWS Certified Cloud Practitioner

Dev Assist / AI: Claude, GitHub Copilot: code generation, test generation from requirements, debugging, documentation, AI-assisted code migration

Test Strategy: Regression, integration, end-to-end and contract-level validation; root cause analysis; flaky-test reduction; quality gates; Agile/Scrum

PROFESSIONAL EXPERIENCE

Capital One | **Senior Test Automation Engineer (Consultant)** | *Remote* | *June 2022 – Present*

- Designed and built a reusable Playwright test automation framework in TypeScript with Cucumber/BDD and Page Object Model, running in parallel inside Docker-based Jenkins pipelines, and automate UI tests in Selenium with Java.
- Built the API automation layer: an auth helper that obtains OAuth tokens using credentials injected from AWS Secrets Manager, Faker-based test data builders, reusable API client functions and shared utilities, with tests focused only on scenarios and assertions.
- Automate REST APIs with Karate, Playwright and Postman, covering status codes, headers, JSON schema and payload validation, authentication, data synchronization, error handling and edge cases.
- Reduced manual testing effort by 70% and improved defect detection by 30% by replacing manual regression with CI/CD-integrated automation.
- Led the modernization of the legacy Ruby/Watir suite into Playwright, planning the migration and using Claude for AI-assisted conversion with a review checklist so every converted test was validated before merging.
- Design integration and end-to-end tests across Salesforce, Amazon Connect and internal data platforms (BusinessCore, One View), validating data flows between systems with SQL and SOQL.
- Analyze failures and drive root cause resolution with engineering: reduced flaky tests by fixing brittle locators, timing issues and shared test data, and used Salesforce debug logs to catch a recursive update loop before release.
- Tested AWS Lambda functions, validating inputs, outputs and error handling with logs in CloudWatch.
- Define automation standards and quality gates with architects and developers; mentor QA engineers and review their test code.

Centene | **QA Engineer** | *January 2022 – June 2022*

- Tested an AWS data pipeline end to end: executed Lambda functions, verified JSON files in Amazon S3, validated Talend ETL transformations, and reconciled data loaded into Snowflake using SQL; tested malformed and incomplete data cases.

System Soft Technologies LLC | **Test Lead** | *Remote* | *February 2021 – January 2022*

- Built a REST Assured (Java) API automation framework with reusable request/response specifications, TestNG data providers and MongoDB data validation.
- Initiated a reusable Selenium/Java automation framework (Page Object Model, TestNG, Maven, Extent Reports, Log4j) adopted by multiple application teams.
- Led QA as Test Lead: planned Agile testing, prioritized work across the team, mentored testers, and maintained Jenkins CI/CD pipelines.

TGROC | **Software Test Engineer (Volunteer)** | *Rochester, NY* | *September 2018 – February 2021*

- Built Python/Selenium UI and REST API test frameworks with Pytest, page objects, fixtures and data-driven tests.

Earlier Experience | **Lead Software Engineer, Lead Engineer, Senior Software Engineer** | *2004 – 2014*

- Sony Mobile Communications, Sweden (2010–2014): led Java/JUnit automation and integration testing for Android middleware.
- RealNetworks (2009–2010): built a data-driven Selenium WebDriver/Java framework and defined the master test strategy.
- Wipro Technologies (2005–2009) and IIT Madras (2004–2005): developed and tested enterprise web applications with SQL and shell scripting.

INDEPENDENT PROJECT

SlickReceipts — AI Receipt-to-Spreadsheet SaaS | Founder & Developer | github.com/narm2005

- Built the product's test suites: Playwright end-to-end tests with page objects and network-level API mocking, a pytest API suite, and an automated LLM accuracy evaluation, running in GitHub Actions with a scheduled daily smoke test.

EDUCATION & CERTIFICATIONS

- Bachelor of Engineering, Electronics and Communications — Periyar University, India, 2003
- Oracle Certified Java Professional (OCJP), Java SE 8 — 2018 • AWS Certified Cloud Practitioner — 2025